Jiawei Guo

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Education

2022 – 2026 **Ph.D., University of South Carolina** Mechanical Engineering.
Research interest: *Deep-Learnig Based Computer Vision.*
Unserpvised-Learning Based Anomaly Detection.
Edge-GPU Device Development.

2020 – 2021 **M.S., University of Southern California** Electrical Engineering.
Research interest: *Computer-Vision Based Brain MRI Analysis*.

2015 – 2019 **B.S., Tianjin University of Technology and Education** Electrical Engineering.
Thesis title: *Design of Vision-based Driver Behavior Analysis* .

Employment

2026–Present **Lehigh University** Postdoctoral Research Associate.

2022 – 2026 **University of South Carolina** Research Assistant.

Research Experience

- **Unsupervised-Learning Based Anomaly Lamb Wave Detection**.
  1. Developed an unsupervised learning framework based on Transformer networks and introduced a novel architecture—termed the Adversarial Transformer (AT)—designed to analyze Lamb wave patterns and detect anomalies.
  2. Developed a comprehensive data analysis pipeline that integrated the proposed model with the data collection sensor, enabling efficient processing and analysis of Lamb wave patterns.
  3. Developed a efficient training pipeline that enables model retraining with newly collected data, ensuring adaptability and continuous improvement in analysis accuracy.

  Funding and Collaboration: This research was sponsored by the US Department of Energy (DOE), in collaboration with Savannah River National Laboratory.

- **Real-Time Rail track Crack Inspection System**.
  1. Developed a transfer learning-based Convolutional Neural Network (CNN) framework for real-time rail track crack inspection, optimizing the model training process for greater efficiency and adaptability across multiple tasks. This framework enables precise segmentation and localization of cracks in real time, enhancing the system's robustness and generalization capability.
  2. Developed a real-time system by seamlessly integrating the model into the edge-computing device, Nvidia AGX Orin, enabling high-speed, on-device processing for rail track crack inspection. This system ensures efficient, real-time detection and analysis, reducing latency and enhancing operational reliability.

  Funding and Collaboration: This research is funded by the Federal Railroad Administration (FRA).

Research Experience (continued)

- **Real-Time Railroad Components Inspection System**.
  1. Developed an efficient vision-based knowledge distillation system for real-time railroad component inspection, optimizing the model size to be lightweight while maintaining high processing speed and accuracy. This system enables precise localization and counting of components, ensuring fast and reliable performance in real-world applications.
  2. Developed a real-time processing pipeline that integrates a high-frame-rate camera, an lightweight model, and the edge-computing device Nvidia AGX Orin, enabling efficient real-time data acquisition, processing, and analysis for enhanced performance in railroad component inspection.

Funding and Collaboration: This research is funded by FRA.

Research Publications

Journal Articles

1. **J. Guo**, Y. Cao, G. Song, J. Ou, and Y. Wang, "A hybrid transformer deep q-network for path planning," *Neurocomputing*, vol. 696, p. 134 135, 2026.
2. J. Ou, G. Song, **J. Guo**, Y. Cao, and Y. Wang, "Gpu-enabled decentralized, multi-robot path planning based on global evolutionary dynamic programming and local particle swarm optimization," *Expert Systems with Applications*, vol. 321, p. 132 321, 2026.
3. **J. Guo**, Y. Qian, and Y. Wang, "A transfer-learning tnet for multitask railroad inspection," *Computers and Electrical Engineering*, vol. 127, p. 110 606, 2025.
4. **J. Guo**, G. Song, and Y. Wang, "A memory and retrieval transformer-based unsupervised learning model for anomaly detection and segmentation," *Pattern Recognition*, p. 113 004, 2025.
5. **J. Guo**, S. Zhang, N. Amiri, L. Yu, and Y. Wang, "An adversarial transformer for anomalous lamb wave pattern detection," *Neural Networks*, vol. 185, p. 107 153, 2025.
6. G. Song, **J. Guo**, Y. Zhang, Y. Qian, and Y. Wang, "An auto-caption-guided hierarchical vision graph transformer for remote sensing image segmentation," *ISPRS Journal of Photogrammetry and Remote Sensing*, 2025, (Under Review).
7. X. Wei, D. Ma, **J. Guo**, *et al.*, "Narrowing signal distribution by adamantane derivatization for amino acid identification using an α-hemolysin nanopore," *Nano Letters*, vol. 24, no. 5, pp. 1494–1501, 2024.
8. **J. Guo**, S. Zhang, Y. Qian, and Y. Wang, "An adaptively weighted loss-enabled lightweight teacher–student model for real-time railroad inspection on edge devices," *Neural Computing and Applications*, vol. 35, no. 34, pp. 24 455–24 472, 2023.
9. Y. Ren, L. Zhang, **J. Guo**, *et al.*, "A theoretical energy loss calculation method for transformer district based on improved convolutional neural networks," *IEEE Transactions on Industry Applications*, vol. 58, no. 5, pp. 5885–5894, 2022.

Conference Proceedings

1. A. Campbell, **J. Guo**, B. Chen, *et al.*, "Inspection of cladding coating with wideband laser induced guided ultrasonic waves," in *TopFuel 2025*, (Accepted), 2025.
2. B. Chen, **J. Guo**, Z. Qian, and Y. Wang, "An autonomous multimodal system for intelligent railway inspection," in *Annual Conference of the PHM Society*, vol. 17, 2025.
3. **J. Guo**, B. Chen, Q. Yu, and Y. Wang, "Railnet: A lightweight transfer learning model for real-time rail component detection and defect segmentation," in *Annual Conference of the PHM Society*, vol. 17, 2025.

4. **J. Guo**, B. Chen, S. Zhang, *et al.*, "A dual-contrastive-attention transformer for unsupervised lamb wave defect detection," in *Annual Conference of the PHM Society*, vol. 17, 2025.
5. **J. Guo**, S. Zhang, Y. Qian, and Y. Wang, "A nanodet model with adaptively weighted loss for real-time railroad inspection," in *Annual Conference of the PHM Society*, vol. 15, 2023.
6. **J. Guo** and S. Wu, "Damage analysis on post-hurricane images taken by satellite based on multiple variants of convolutional neural networks," in *2021 2nd International Conference on Big Data & Artificial Intelligence & Software Engineering (ICBASE)*, IEEE, 2021, pp. 515–518.

Patents

1. Y. Wang, **J. Guo**, D. Cahl, and C. Papageorgiou, "Configurable, platform-independent computer vision module for coordinated infrastructure inspection," 63/840955, Provisional Patent Application, filed July 9, 2025.

Skills

Languages	English, Mandarin Chinese.
Coding	Python, C++, MATLAB, Java, LaTeX.
Tools	Anaconda, Pytorch, Linux, Numpy, Pandas, VScode, PyCharm.
Knowledge	Deep Learning, Machine Learning, Computer Vision, Reinforcement Learning, Probability, Linear Algebra.
Hardware	Nvidia Jetson AGX Xavier, Nvidia Jetson AGX Orin.